Double-Angle Formulas and Trigonometric Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Double-angle formulas**

Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.

$\cos^2 x = \frac{1+\cos 2x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$.

2 **[Foundational]** • **Target: Solving trigonometric equations**

Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.

$\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$.

3 **[Foundational]** • **Target: Half-angle formulas**

Use a half-angle formula to find the exact value of $\cos 15°$ (using $15° = \frac{30°}{2}$).

$\cos 15° = \sqrt{\frac{1+\cos 30°}{2}} = \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$ (positive since $15°$ is in Quadrant I).

4 **[Foundational]** • **Target: Half-angle formulas**

Given $\cos\theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin\frac{\theta}{2}$ exactly.

$\sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = \sqrt{\frac{18/25}{2}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$ (positive since $\frac{\theta}{2}$ is in Quadrant I).

5 **[Foundational]** • **Target: Equations requiring double-angle substitution**

Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

$1 - 2\sin^2 x + \sin x = 0$, so $2\sin^2 x - \sin x - 1 = 0$: $(2\sin x + 1)(\sin x - 1) = 0$. From $\sin x = -\frac{1}{2}$: $x = \frac{7\pi}{6}, \frac{11\pi}{6}$; from $\sin x = 1$: $x = \frac{\pi}{2}$. Solution set: $\left\{\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\right\}$.

6 **[Foundational]** • **Target: Applications of double-angle formulas**

A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45°$.

$R(\theta) = \frac{v^2}{g}\sin 2\theta$ is maximized when $\sin 2\theta = 1$, i.e. $2\theta = 90°$, so $\theta = 45°$.

7 **[Foundational]** • **Target: Applications of double-angle formulas**

Given $\sin\theta = \frac{1}{3}$ with θ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

$\cos\theta = \sqrt{1 - 1/9} = \frac{2\sqrt{2}}{3}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{2}{9} = \frac{7}{9}$. $\sin 2\theta = 2\sin\theta\cos\theta = 2 \cdot \frac{1}{3} \cdot \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{9}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Half-angle formulas for tangent**

State the half-angle formula for $\tan\frac{\theta}{2}$ in terms of $\sin\theta$ and $\cos\theta$ (the form avoiding a square root), and use it to find $\tan 15°$.

$\tan\frac{\theta}{2} = \frac{\sin\theta}{1+\cos\theta}$. With $\theta = 30°$: $\tan 15° = \frac{\sin 30°}{1+\cos 30°} = \frac{1/2}{1+\sqrt{3}/2} = \frac{1}{2+\sqrt{3}} = 2-\sqrt{3}$ (after rationalizing).

9 **[Intermediate]** • **Target: Equations with multiple angle types**

Solve $\cos 2x = 1 - 3\sin x$ on $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

$1 - 2\sin^2 x = 1 - 3\sin x \Rightarrow -2\sin^2 x + 3\sin x = 0 \Rightarrow \sin x(3 - 2\sin x) = 0$. From $\sin x = 0$: $x = 0, \pi$. From $\sin x = \frac{3}{2}$: no solution (out of range). Solution set: $\{0, \pi\}$.

10 **[Intermediate]** • **Target: Triple-angle preview**

Using $\sin(2x + x) = \sin 2x\cos x + \cos 2x\sin x$ together with the double-angle formulas, derive the triple-angle formula $\sin 3x = 3\sin x - 4\sin^3 x$.

$\sin 3x = \sin 2x\cos x + \cos 2x\sin x = (2\sin x\cos x)\cos x + (1 - 2\sin^2 x)\sin x = 2\sin x\cos^2 x + \sin x - 2\sin^3 x$. Substitute $\cos^2 x = 1 - \sin^2 x$:

$= 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x = 2\sin x - 2\sin^3 x + \sin x - 2\sin^3 x = 3\sin x - 4\sin^3 x$.

11 **[Intermediate]** • **Target: Graphical verification of a double-angle identity**

The functions $y = \cos 2x$ and $y = 1 - 2\sin^2 x$ are graphed together below. Explain what it means that the two curves appear identical, and state the identity this confirms.

The two curves overlapping perfectly (indistinguishable) is a visual confirmation of the identity $\cos 2x = 1 - 2\sin^2 x$ — since both functions produce the exact same output for every input x, their graphs coincide everywhere.

12 **[Intermediate]** • **Target: Solving equations restricted to a smaller interval**

Solve $2\cos 2x - 1 = 0$ for $x \in [0, \pi)$ only (note the restricted domain).

$\cos 2x = \frac{1}{2}$: over $[0, 2\pi)$ for $2x$, solutions are $2x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$, giving $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$. Restricting to $x \in [0, \pi)$: only $x = \frac{\pi}{6}, \frac{5\pi}{6}$ qualify.

13 **[Intermediate]** • **Target: Solving trigonometric equations**

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x + 6)] = 3$. Convert to exponential form: $x(x + 6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic: $(x + 9)(x - 3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

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14 [Intermediate] • Target: Half-angle formulas

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Step 1: Identify ellipse parameters: The vertices and foci lie on the x-axis, so the major axis is horizontal and centered at $(0, 0)$. Here semi-major axis $a = 5$ and focal distance $c = 3$. Step 2: Compute semi-minor axis b: For an ellipse: $b^2 = a^2 - c^2 = 5^2 - 3^2 = 25 - 9 = 16 \Rightarrow b = 4$. Step 3: Formulate equation and invariants: $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Major axis length $2a = 10$, minor axis length $2b = 8$, and eccentricity $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Equations requiring double-angle substitution • Cross-Lesson: 8.03 Introduction to Limits

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

The vertices share $x = 1$, so the major axis is vertical. Center is the midpoint of the vertices: $(h, k) = \left(1, \frac{6 + (-4)}{2}\right) = (1, 1)$. Semi-major axis: $a = 6 - 1 = 5$ (so $a^2 = 25$). Focal distance from center to $(1, 4)$: $c = 4 - 1 = 3$. For an ellipse, $b^2 = a^2 - c^2 = 25 - 9 = 16 \Rightarrow b = 4$. Standard equation: $\frac{(x-1)^2}{16} + \frac{(y-1)^2}{25} = 1$. Eccentricity: $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

16 [Challenging] • Target: Applications of double-angle formulas • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

Complete the square on the y-terms: $y^2 - 4y = -8x - 28 \Rightarrow (y^2 - 4y + 4) = -8x - 28 + 4$.
$(y - 2)^2 = -8x - 24 = -8(x + 3)$. This matches $(y - k)^2 = 4p(x - h)$ with:Vertex: $(h, k) = (-3, 2)$
Focal parameter: $4p = -8 \Rightarrow p = -2$ (opens horizontally to the left)Focus:
$(h + p, k) = (-3 + (-2), 2) = (-5, 2)$Directrix: $x = h - p = -3 - (-2) \Rightarrow x = -1$

17 [Challenging] • Target: More equation practice • Cross-Lesson: 9.02 Geometric Sequences and Series

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

(a) Half-life calculation: $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693147}{0.035} \approx 19.8\ \text{years}$. (b) Remaining mass at $t = 40$:

$A(t) = A_0 e^{-kt} = 250e^{-0.035(40)} = 250e^{-1.4} \approx 250(0.246597) \approx 61.65\ \text{mg}$.

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18 [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 9.03 The Binomial Theorem

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 1.01 Functions and Function Notation

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Triple-angle preview • Cross-Lesson: 1.02 Domain and Range

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Double-angle formulas

Find the half-angle formula for $\cos(\theta/2)$.

a. $\pm\sqrt{\frac{1+\cos\theta}{2}}$ ✓

b. $\cos\theta/2$

c. $\pm\sqrt{\frac{1-\cos\theta}{2}}$

d. $\sqrt{1+\cos\theta}$

Applying Trigonometric Functions principles, Half-angle formula for cosine. Therefore, the verified correct option is $\pm\sqrt{\frac{1+\cos\theta}{2}}$.

22 [Foundational] • Target: Solving trigonometric equations

State the double-angle formula $\sin 2\theta =$

a. $2\cos^2\theta - 1$

b. $\sin^2\theta - \cos^2\theta$

c. $2\sin\theta\cos\theta$ ✓

d. $\cos^2\theta - \sin^2\theta$

Applying Trigonometric Functions principles, Double angle formula. Therefore, the verified correct option is $2\sin\theta\cos\theta$.

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23 [Foundational] • Target: Half-angle formulas

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.

a. 2 ✓

b. 6

c. -4

d. $\frac{2}{3}$

To find the gradient of the tangent to the curve at $P(3, 2)$, we determine the derivative $\frac{dy}{dx}$ and evaluate it at $x = 3$. 1. Differentiate $y = x^2 - 4x + 5$ with respect to x using the power rule: $\frac{dy}{dx} = 2x - 4$ 2. Substitute $x = 3$ into the derivative: $\frac{dy}{dx}\Big|_{x=3} = 2(3) - 4 = 6 - 4 = 2$ Common errors: - 6 occurs if only the first term is differentiated (giving $2x$) and evaluated at $x = 3$, omitting the deriv.

24 [Foundational] • Target: Equations requiring double-angle substitution

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$ ✓

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on C_2, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at A is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

25 [Foundational] • Target: Applications of double-angle formulas

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e}-2e-1}{2e}$ ✓

b. $\frac{4\sqrt{e}-2e+1}{2e}$

c. $\frac{2e-3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e}-2e-1}{2e}$

1. **Find the stationary point M:** Using the quotient rule to differentiate $y = \frac{\ln x}{x^2}$: $\frac{dy}{dx} = \frac{\frac{1}{x}\cdot x^2 - (\ln x)(2x)}{x^4} = \frac{x(1-2\ln x)}{x^4} = \frac{1-2\ln x}{x^3}$. Setting $\frac{dy}{dx} = 0$: $1 - 2\ln x = 0 \Rightarrow \ln x = \frac{1}{2} \Rightarrow x = e^{1/2} = \sqrt{e}$. The y-coordinate at M is: $y = \frac{\ln(\sqrt{e})}{(\sqrt{e})^2} = \frac{\frac{1}{2}}{e} = \frac{1}{2e}$. Since M is a stationary point, the tangent line L is horizontal wit.

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26 [Foundational] • Target: More equation practice

Consider the system of linear equations represented by the matrix equation $(1, 1, 1; 1, 2, 3; 3, 5, k)(x; y; z) = (6; 14; m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x, y, z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2 = x^2 + y^2 + z^2$. What is the value of $k + m + d^2$?

a. **55** ✓

b. 47

c. 38

d. 109

We solve the problem in three main stages: 1. **Determine the parameter k:** For the system to have infinitely many solutions, the coefficient matrix must be singular (its determinant must be zero): $\det(1, 1, 1; 1, 2, 3; 3, 5, k) = 1(2k - 15) - 1(k - 9) + 1(5 - 6) = k - 7$ Setting $k - 7 = 0$ yields $k = 7$. 2. **Determine the parameter m:** Set up the augmented matrix with $k = 7$ and perform Gaussian elimination.

27 [Foundational] • Target: Half-angle formulas for tangent

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k - y}{2x + 1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1 ✓

b. 11

c. $\frac{5}{4}$

d. 5

To find the constant k, we solve the differential equation by separation of variables: $\int \frac{1}{k - y}\,dy = \int \frac{1}{2x + 1}\,dx$ Integrating both sides (paying close attention to the chain rule): $-\ln|k - y| = \frac{1}{2}\ln(2x + 1) + C$ Multiplying by -1 and exponentiating: $\ln|k - y| = -\frac{1}{2}\ln(2x + 1) - C = \ln\left((2x + 1)^{-1/2}\right) - C$
$|k - y| = \frac{A}{\sqrt{2x + 1}}$ where $A = e^{-C} > 0$ Removing the absolute value gives th.

28 [Foundational] • Target: Equations with multiple angle types

For a constant k, the equation $\sqrt{2x + k} = x - 1$ has $x = 5$ as a valid solution. What is the extraneous solution produced when solving this equation by squaring both sides?

a. -5

b. -1 ✓

c. 1

d. 6

We solve the problem in two main steps: 1. **Find the parameter k:** Since $x = 5$ is a solution to the original radical equation, substitute $x = 5$ into the equation: $\sqrt{2(5) + k} = 5 - 1$ $\sqrt{10 + k} = 4$ Squaring both sides gives: $10 + k = 16 \Rightarrow k = 6$ 2. **Solve the equation and find the extraneous solution:** Substitute $k = 6$ back into the equation: $\sqrt{2x + 6} = x - 1$ Square both sides to eliminate the radical: $2x + 6 = (x - 1)^2$ $2x + 6 = x^2 - 2x + 1$ Rearrange all terms to one s.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Triple-angle preview**

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2 ✓

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate y with respect to x using the chain rule. Let $u = 2x^3 + 1$, so $y = \ln(u)$. Then: $\frac{du}{dx} = 6x^2$ and $\frac{dy}{du} = \frac{1}{u}$ Applying the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3 + 1}$ Now evaluate the derivative at $x = 1$.

30 **[Intermediate]** • **Target: Graphical verification of a double-angle identity**

Consider the equation $\sqrt{2x + k} = x - 1$, where k is a constant. If $x = 5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

a. -1 ✓

b. 1

c. -5

d. 6

Step 1: Find the value of the parameter k by substituting the valid solution $x = 5$ into the equation: $\sqrt{2(5) + k} = 5 - 1 \Rightarrow \sqrt{10 + k} = 4$ Squaring both sides gives $10 + k = 16$, so $k = 6$. Step 2: Substitute $k = 6$ back into the equation and solve by squaring both sides: $\sqrt{2x + 6} = x - 1$
$2x + 6 = (x - 1)^2 = x^2 - 2x + 1$ $x^2 - 4x - 5 = 0$ Factoring the quadratic equation gives:
$(x - 5)(x + 1) = 0 \Rightarrow x = 5$ or $x = -1$ Step 3: Check for extraneous sol.

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31 [Intermediate] • Target: Solving equations restricted to a smaller interval

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$ ✓

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

To solve the differential equation, we use separation of variables: 1. **Separate the variables**: $e^y\,dy = 2x\,dx$ 2. **Integrate both sides**: $\int e^y\,dy = \int 2x\,dx$ $e^y = x^2 + C$ 3. **Apply the initial condition** $y = 0$ when $x = 0$: $e^0 = 0^2 + C \Rightarrow 1 = C$ *(A frequent error is assuming $C = 0$ because the boundary values are zero, which leads to the incorrect solution $y = \ln(x^2)$.)* 4. **Express y explicitly in terms of x**.

32 [Intermediate] • Target: Double-angle formulas

State the angle addition formula $\sin(A + B) =$

a. $\cos A\cos B - \sin A\sin B$

b. $\sin A\cos B + \cos A\sin B$ ✓

c. $\sin A\cos B - \cos A\sin B$

d. $\sin A + \sin B$

Applying Trigonometric Functions principles, Angle addition formula. Therefore, the verified correct option is $\sin A\cos B + \cos A\sin B$.

33 [Intermediate] • Target: Solving trigonometric equations

State the cofunction identity $\sin(90^\circ - \theta) =$

a. $\tan\theta$

b. $\cos\theta$ ✓

c. $-\cos\theta$

d. $\sin\theta$

Applying Trigonometric Functions principles, Cofunction identity. Therefore, the verified correct option is $\cos\theta$.

34 [Intermediate] • Target: Half-angle formulas

Which differential equation correctly models the statement: "The rate of decrease of the temperature T of an object is proportional to the difference between T and the surrounding temperature T_e."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$ ✓

c. $\frac{dN}{dt} = kN(S - N)$

d. $\frac{dm}{dt} = -km$

The correct model is $\frac{dT}{dt} = -k(T - T_e)$, where $k > 0$ is a constant of proportionality.

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35 **[Intermediate]** • **Target: Equations requiring double-angle substitution**

Which differential equation correctly models the statement: "The rate of increase of the number of infected people N is proportional to the product of N and the number of remaining susceptible people $(S - N)$, for fixed population S."?

a. $\frac{dm}{dt} = -km$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dT}{dt} = -k(T - T_e)$

d. $\frac{dN}{dt} = kN(S - N)$ ✓

The correct model is $\frac{dN}{dt} = kN(S - N)$, where $k > 0$ is a constant of proportionality.

36 **[Intermediate]** • **Target: Applications of double-angle formulas**

Which differential equation correctly models the statement: "The rate of increase of the volume V of a balloon is proportional to the square of its radius r."?

a. $\frac{dT}{dt} = -k(T - T_e)$

b. $\frac{dm}{dt} = -km$

c. $\frac{dV}{dt} = kr^2$ ✓

d. $\frac{dP}{dt} = kP$

The correct model is $\frac{dV}{dt} = kr^2$, where $k > 0$ is a constant of proportionality.

37 **[Intermediate]** • **Target: More equation practice**

Which differential equation correctly models the statement: "The rate of decrease of a radioactive mass m is proportional to m."?

a. $\frac{dN}{dt} = kN(S - N)$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dP}{dt} = kP$

d. $\frac{dm}{dt} = -km$ ✓

The correct model is $\frac{dm}{dt} = -km$, where $k > 0$ is a constant of proportionality.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 9.03 The Binomial Theorem

Which differential equation correctly models the statement: "The rate of increase of a population P is proportional to P itself."?

a. $\frac{dP}{dt} = kP$ ✓

b. $\frac{dT}{dt} = -k(T - T_e)$

c. $\frac{dm}{dt} = -km$

d. $\frac{dN}{dt} = kN(S - N)$

The correct model is $\frac{dP}{dt} = kP$, where $k > 0$ is a constant of proportionality.

39 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 1.01 Functions and Function Notation

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at $(3, 4)$.

a. $3x + 4y = 25$ ✓

b. $x + y = 7$

c. $4x + 3y = 25$

d. $3x - 4y = 25$

Tangent at (x_1, y_1): $x_1x + y_1y = r^2 \Rightarrow 3x + 4y = 25$.

40 [Challenging] • Target: Triple-angle preview • Cross-Lesson: 1.02 Domain and Range

Find the area of a sector with radius 4 and central angle $\pi/3$.

a. 2π

b. $\frac{4\pi}{3}$

c. $\frac{8\pi}{3}$ ✓

d. 4π

Area $= \frac{1}{2}r^2\theta = \frac{1}{2}(16)(\pi/3) = 8\pi/3$.

41 [Challenging] • Target: Graphical verification of a double-angle identity • Cross-Lesson: 1.03 Average Rate of Change

A radioactive substance has half-life 2 years and initial mass 16g. Find the mass remaining after 4 years.

a. **4** ✓

b. 16

c. 8

d. 5

After 2 half-life period(s) (4 years), mass $= 16/2^2 = 4$g.

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42 [Challenging] • Target: Solving equations restricted to a smaller interval • Cross-Lesson: 2.01 Right-Triangle Trigonometry

State the Weierstrass substitution $t = \tan(x/2)$.

a. $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2\,dt}{1+t^2}$ ✓

b. $x = 2\arctan t$ only

c. $\tan x = 2t$

d. $\sin x = t$, $\cos x = \sqrt{1-t^2}$

Standard half-angle (Weierstrass) substitution.

43 [Challenging] • Target: Double-angle formulas • Cross-Lesson: 2.02 Radian Measure and Arc Length

A system with more unknowns than equations is called:

a. Consistent

b. Overdetermined

c. Trivial

d. **Underdetermined** ✓

Fewer equations than unknowns: underdetermined.

44 [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Derive the Mollweide's equation: $\frac{a+b}{c} = \frac{\cos\frac{A-B}{2}}{\sin C/2}$ for triangle ABC.

a. Only for right triangles

b. Use the law of cosines

c. Cannot be derived

d. **Use the law of sines and angle sum property** ✓

Standard derivation using law of sines.

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45 [Challenging] • Target: Half-angle formulas • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the equation of the tangent to the ellipse $\frac{x^2}{4} + y^2 = 1$ at $(1, 32)$.

a. $\frac{x}{4} - y = 1$

b. $\frac{x}{4} + \frac{\sqrt{3}y}{2} = 1$ ✓

c. $x + y = 2$

d. $x + y = 1$

Tangent: $xx_1 4 + yy_1 = 1x4 + 32y = 1$.

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